

Telecommunication Customer Churn Prediction

AI & ML Capstone Project - Skyrec Academy

Name: P.G.R.Hasith Pusswella

Index Number: SKML25874

1. Project Idea & Overview

In the telecommunication and networking industry, customer retention is highly critical for maintaining profitability. Customer churn (when a subscriber leaves the network for a competitor) leads to significant revenue loss.

The objective of this project is to build an end-to-end Machine Learning web application that predicts the likelihood of a customer churning based on their usage patterns and account details. By analyzing features such as "Tenure", "Monthly Charges", "Contract Type", and "Internet Service", this system allows telecom operators to proactively identify high-risk customers and offer targeted retention promotions.

2. Final Model Performance

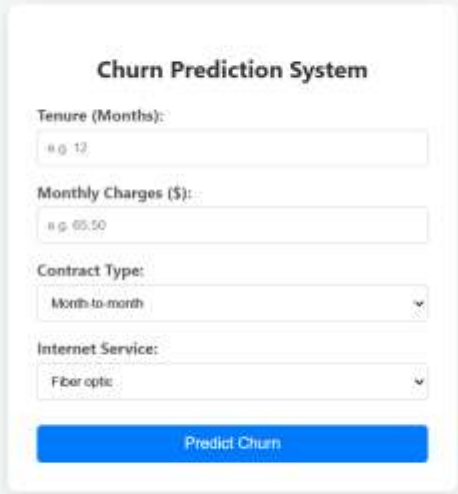
The predictive engine of this system is powered by a Logistic Regression model. After rigorous data preprocessing, encoding, and scaling, the model was evaluated using a testing dataset.

The final model achieved an **overall accuracy of 81%**, demonstrating strong predictive capabilities for real-world scenarios. Key performance metrics include:

- **Non-Churn (Class 0):** F1-Score of 0.87, indicating the model is highly accurate at identifying loyal customers.
- **Churn (Class 1):** Precision of 71%, meaning when the model flags a customer as high-risk, it is correct 71% of the time.

The accompanying Confusion Matrix confirms that the model successfully balances the detection of actual churners while keeping false positives at a manageable level.

3. System Screenshots



The screenshot displays the 'Churn Prediction System' web interface. It features a white card with a light blue background. The card has a title 'Churn Prediction System' in bold. Below the title, there are four input fields: 'Tenure (Months):' with a text input containing 'e.g. 12', 'Monthly Charges (\$):' with a text input containing 'e.g. 65.50', 'Contract Type:' with a dropdown menu showing 'Month-to-month', and 'Internet Service:' with a dropdown menu showing 'Fiber optic'. At the bottom of the card is a blue button labeled 'Predict Churn'.

Frontend UI before prediction.

The image shows a web form titled "Churn Prediction System". It contains four input fields: "Tenure (Months)" with the value "4", "Monthly Charges (\$)" with the value "9", "Contract Type" with the value "Month-to-month", and "Internet Service" with the value "Fiber optic". Below these fields is a blue button labeled "Predict Churn". At the bottom, a red message box states "Yes, this customer is likely to Churn."

Churn Prediction System

Tenure (Months):
4

Monthly Charges (\$):
9

Contract Type:
Month-to-month

Internet Service:
Fiber optic

Predict Churn

Yes, this customer is likely to Churn.

Frontend UI after a successful prediction -01

The image shows a web form titled "Churn Prediction System". It contains four input fields: "Tenure (Months)" with the value "10", "Monthly Charges (\$)" with the value "20", "Contract Type" with the value "One year", and "Internet Service" with the value "Fiber optic". Below these fields is a blue button labeled "Predict Churn". At the bottom, a green message box states "No, this customer is likely to stay."

Churn Prediction System

Tenure (Months):
10

Monthly Charges (\$):
20

Contract Type:
One year

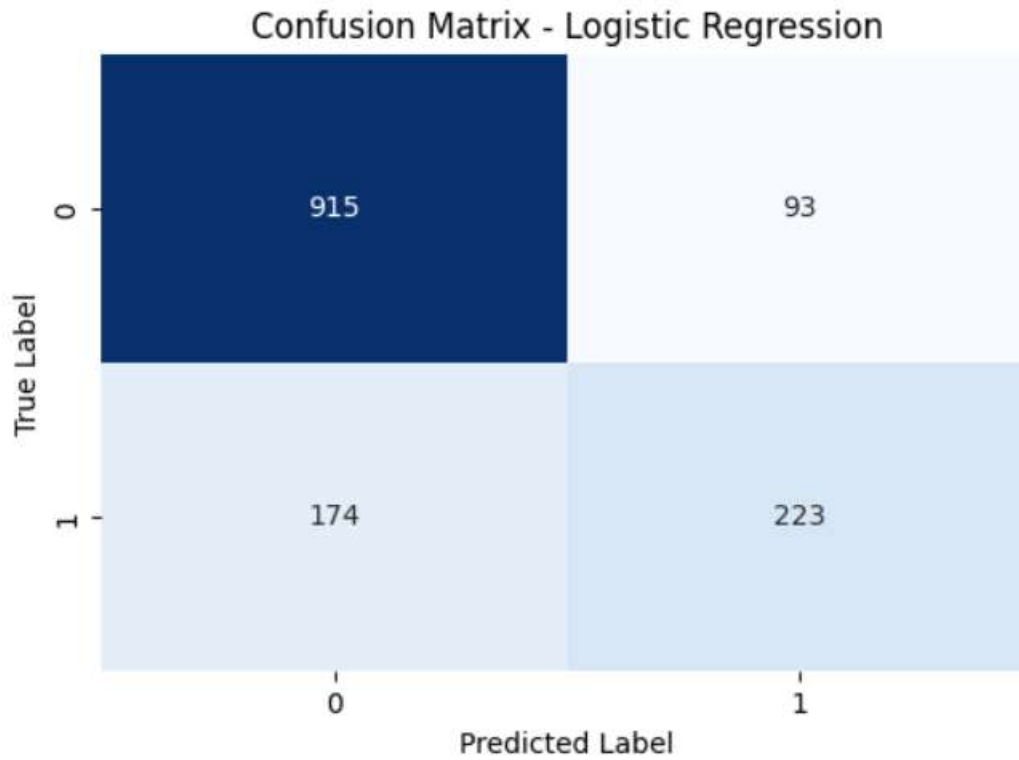
Internet Service:
Fiber optic

Predict Churn

No, this customer is likely to stay.

Frontend UI after a successful prediction -02

Confusion Matrix Plot:



Confusion Matrix

Starting Model Training...

Logistic Regression Accuracy: 0.8100

Decision Tree Accuracy: 0.7331

Random Forest Accuracy: 0.7986

🏆 Best Model Selected: Logistic Regression with an Accuracy of 0.8100

Model Accuracy